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Selfie-posting on social networking sites, body surveillance, and exercise for weight control among Chinese young women with low body mass index

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ABSTRACT

Under the perspective of self-perception theory and objectification theory, this study aims to examine the association between selfie-posting on social networking sites (SNSs) and exercise for weight control among Chinese young women, as well as the mediating role of body surveillance. A sample of 467 Chinese female students with a BMI lower than 24 were recruited. Participants voluntarily completed questionnaires assessing their selfie-posting behavior on SNSs, body surveillance, and their weight control motivation for exercise. The results show that selfie-posting on SNSs is positively associated with exercise for weight control, and that body surveillance significantly mediates this association. The results suggest that interventions targeting selfie behaviors on SNSs may be an effective way to prevent unhealthy exercise motivations, such as exercise for weight control, among young women with a low BMI.

1. Introduction

In the present day, many young women are motivated to both obtain and maintain their perceived ideal body shape, as thin and fit physiques have become the idealized body standard among women (Sabiston, Crocker, & Munroe-Chandler, 2005). Exercise has always been one of the most common concomitant behaviors for women in terms of managing their weight and improving their appearance (Buchholz & Crowther, 2014), and weight control (a set of appearance-based motivations) has, subsequently, become a major motivation driving women to exercise (Hurst, Dittmar, Banerjee, & Bond, 2017 Kilpatrick, Hebert, & Bartholomew, 2005). But for women with a low body mass index (BMI) (BMI < 24 kg/m^2, including those women who are normal weight and underweight), exercising for weight loss purposes may be unnecessary (Liechty, 2010), yet remains common. According to Liechty's (2010) study on 5173 non-overweight adolescent girls in the United States, 63.6% of the participants use exercise as a means to lose weight. A survey of 7662 Chinese women similarly found that 67.5% of those in the normal and underweight categories believe that they need to lose weight or get in to shape, and that exercise is the main way they use in order to achieve this (39 Health Network, 2013).

In addition, relevant studies further examined the influences of exercise for weight control and have found that it is generally correlated with negative outcomes, including: the development and maintenance of eating problems and disorders (Gonçalves & Gomes, 2012), depressive symptoms, lower self-esteem (DiBartolo, Lin, Montoya, Neal, & Shaffer, 2007), and a poor body image (Vartanian, Wharton, & Greeen, 2012), especially among those with a low BMI. Thus, to some extent, exercise for weight control can, therefore, be regarded as an unhealthy exercise motivation that may be accompanied by undesirable and deleterious influences (DiBartolo et al., 2007). Therefore, considering the prevalence of exercise by women with a low BMI who are attempting to lose weight, and the harmfulness of exercise for weight control, it is crucial to examine why young women with a low BMI use exercise primarily for weight control.

Against this background, the factors underlying the use of exercise for weight control have been widely examined, with media use being found to be closely associated with it (Pearl, Dovidio, Puhl, & Brownell,

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2015). In the present day, social networking sites (SNSs) are widely used, often acting as conduits for popular media. WeChat Moments and Qzone are the most popular SNSs in China, with their users representing 83.4% and 58.8% of Chinese Internet users, respectively (CNNIC, 2018). More than half of these SNS users are young people under the age of 29 (CNNIC, 2017). Additionally, SNSs are also unique in their dissemination of user-created content and selective self-presentation (Dhir, Pallesen, Torsheim, & Andreassen, 2016); the most common user-created content are selfies (Mills, Musto, Williams, & Tiggemann, 2018).

Selfie-posting is an appearance-focused "photo activity" on SNSs that is very common among young women (Yellowlees, Dingemans, Veldhuis, & de Vaate, 2019). Studies examining the outcomes of selfie-posting on SNSs have found that the activity is closely associated with body image issues — including body dissatisfaction, drive for thinness, and eating disorders (Cohen, Newton-John, & Slater, 2018 Mills et al., 2018). Although there is no direct evidence of the relationship between selfie-posting and exercise motivation, the image focus of photo activity on SNSs is most likely to maximize appearance motivations for exercise (Fatt, Fardouly, & Rapee, 2019). Therefore, this study aims to investigate the association between selfie-posting on SNSs and weight control motivation for exercise among Chinese young women with a low BMI, with one potential mediating mechanism underlying this association.

1.1. Selfie-posting on SNSs and exercise for weight control

A selfie refers to a self-portrait image taken by oneself, usually via a smartphone, often to be posted on SNSs (Qiu, Lu, Yang, Qu, & Zhu, 2015). To present the most attractive version of themselves, users tend to carefully select and edit their selfies before posting (Zheng, Ni, & Luo, 2019). This process usually leads to a focus on body image, appearance, and weight (De Vries, Peter, Nikken, & De Graaf, 2014), further motivating individuals to improve their appearance and to control their weight. According to self-perception theory, the characteristics or traits one presents to others become more salient and are more likely to guide an individual's future behaviors (Schienker, Dlugolecki, & Doherty, 1994). This theory has been used to explain the relationship between self-presentation (selfie-posting) on SNSs and restrained eating (another common way of weight control other than exercise) (Niu et al., 2019). Empirically, studies show that women who post and manipulate their selfies are more likely to be dissatisfied with their bodies (McLean et al., 2015a McLean et al., 2015b) and have a greater desire for cosmetic surgery and dietary restraint (De Vries et al., 2014 McLean et al., 2015b). Meanwhile, appearance-focused images which are posted on SNSs are found to be associated with appearance-based exercise motivation (Fatt et al., 2019) and maladaptive exercise behaviors (Holland & Tiggemann, 2017). Based on the above discussions, it is hypothesized that selfie-posting on SNSs is positively associated with exercise for weight control among young women with a low BMI (Hypothesis 1).

1.2. The mediating role of body surveillance

In addition, selfie-posting may also be associated with body surveillance (Butkowski, Dixon, & Weeks, 2019), which is an important risk factor for body image disturbances and related issues (Moradi & Huang, 2008). Body surveillance refers to persistent thinking and the habitual monitoring of one's body as an outside observer (McKinley & Hyde, 1996). It is the cognitive component of objectified body consciousness and the behavioral manifestation of self-objectification. Young women usually have a higher degree, and frequency, of self-objectification (Grabe & Jackson, 2009) — a trait which holds true among Chinese women (Shen, 2018). As selfies are always both self-focused and appearance-focused, a state of objective self-awareness can be activated by focusing exclusively on oneself and one's appearance (Yellowlees et al., 2019). Moreover, selfies posted on SNSs may receive appearance-based feedback (comments and "likes") from others (Cohen et al., 2018), both of which can make women scrutinize their own body from an observer's perspective. Thus, in accordance with objectification theory (Fredrickson & Roberts, 1997), selfie-posting may foster body surveillance behavior via the act of focusing on one's own body image (Cohen et al., 2018). Further empirical studies demonstrate that selfie-posting is associated with adolescent girls' self-objectification (Zheng et al., 2019).

Furthermore, women who constantly monitor their bodies not only tend to put an extreme emphasis on outward appearance and weight, but are also more likely to be motivated in adopting behaviors (such as exercise) to obtain a body that meets society's perceived expectations; in other words, they are more likely to exercise for the purpose of weight control (Strelan, Mehaffey, & Tiggemann, 2003). Additional studies identify that self-objectification is positively correlated with exercise for weight control (Strelan et al., 2003), and that body surveillance can play a mediating role in the relationship between selfie activity and the drive for thinness among young women (Butkowski et al., 2019). Therefore, based on objectification theory, along with previous research, it is hypothesized that body surveillance would mediate the relationship between selfie-posting on SNSs and exercise for weight control among young women with a low BMI (Hypothesis 2).

In summary, based on self-perception theory and objectification theory, this study aims to investigate the association between selfie-posting on SNSs and exercise for weight control among Chinese young women with a low BMI, as well as the potential mediating role played by body surveillance upon this association. This might help researchers understand the impact of selfie-posting on an individual's psychosocial development and, subsequently, provide better-suited guidance for reasonable SNS use, along with interventions targeting unhealthy exercise motivations among young women with a low BMI.

2. Method

2.1. Participants

After obtaining approval from the Academic Committee for Scientific Research at the researchers' university, participants were recruited randomly (through convenience sampling) from a pool of women with active SNS accounts from mental health classes at two universities in central China. In this study, a total of 509 college women voluntarily participated in the pencil-and-paper survey. All participants signed informed consent forms. In order to guarantee the integrity and validity of the collected questionnaires, 31 individuals who had failed to fill in the forms completely, answered regularly, or answered with extreme values were removed. In addition, and according to the Chinese BMI cutoffs, 11 individuals with BMI scores over 24 kg/m^2 (who fall into either the overweight or obese categories) were excluded. The final useable sample is composed of 467 college women with a BMI lower than 24 kg/m^2, including 134 underweight participants (BMI $<$ 18.5) and 333 normal weight participants (18.5 $\leq$ BMI $<$ 24). Participants' ages range between 17 and 23 years old (M = 20.27, SD = 1.31).

2.2. Measurements

2.2.1. Selfie-posting

Five items used by McLean et al., 2015a were adopted in this study in order to measure the frequency of participants' selfie-related behaviors on SNSs — including selfie-taking, selfie-editing, and selfie-posting (e.g., editing and posting selfies on SNSs). Participants were asked to respond to a 5-point Likert scale to each item (1 = *never*, 5 = *very often*). Responses were averaged so as to form a scale score, reflecting the frequency of selfie-related behaviors by participants on SNSs, with higher scores indicating a higher frequency. In the present study, the confirmatory factor analysis indicates a good fit, where: χ^2/df = 0.65, RMSEA = 0.000, CFI = 1.000, SRMR = 0.005; and for which the Cronbach's alpha is 0.82.

2.2.2. Body surveillance

The body surveillance subscale of the Chinese version (Chen, 2014) of the Objectified Body Consciousness Scale (OBCS; Mckinley & Hyde, 1996) was used to assess participants' tendency to monitor their physical appearance. The subscale consisted of 8 items rated on a 7-point Likert scale (1 = *definitely disagree*, 7 = *definitely agree*). Some items were scored in reverse, and all items were summed and averaged for a scale score reflecting body surveillance ; higher scores indicategreater frequency body surveillance. In the present study, the confirmatory factor analysis shows a good fit: $\chi^2/df = 3.35$, RMSEA = 0.070, CFI = 0.948, SRMR = 0.042; the Cronbach's alpha for the subscale is 0.70.

2.2.3. Exercise for weight control

The exercise for weight control subscale of the Exercise Dependence Questionnaire (Ogden, Veale, & Summers, 1997), which consists of 4 items rated on a 7-point Likert scale (1 = *completely disagree*, 7 = *completely agree*) (e.g., I exercise to control my weight) was used in this study. Item responses were summed and averaged to create a single measure representing exercise for weight control, with higher scores indicating more exercise for weight control. In the present study, it was transited into Chinese through a standard procedure to determine the version applicable to the Chinese context. The subscale score is also found to have good validity and reliability — the confirmatory factor analysis indicates a good fit: $\chi^2/df = 0.748$, RMSEA = 0.000, CFI = 1.000, SRMR = 0.007; the Cronbach's alpha for the present sample is 0.83.

2.2.4. Body mass index

Participants' BMI was calculated by using the equation BMI = weight (kg)/height (m)^2 based on their self-reported height and weight. In this study, the BMI of participants ranged between 15.21 and 23.98 (M = 19.66, SD = 1.79).

2.2.5. Covariates

In the following statistical analyses, the present study controls for age, BMI, and the number of friends of participants on SNSs — this is due to the fact that age and BMI are important factors associated with exercise motivation (Azeem & Antony, 2018 Brunet & Sabiston, 2011) and that the number of friends of participants on SNSs (as a measure of engagement with SNSs) could impact the relationship between SNS usage and body image-related issues (Holland & Tiggemann, 2016). To ensure that our results are not confounded by participants' age, BMI or their number of friends on SNSs, we include these variables as covariates.

3. Results

The means, standard deviations, and correlation coefficients for the study variables are presented in Table 1. Selfie-posting, body surveillance, and exercise for weight control are positively correlated in pairs. Thus, Hypothesis 1 is supported.

We subsequently used the PROCESS macro (http://www.afhayes.com) for SPSS (Model 4) — as suggested by Hayes (2013) — in order to test whether *body surveillance* mediates the relationship between selfie-posting and exercise for weight control. The regression results (see Fig. 1) show that selfie-posting is positively associated with exercise for weight control ($\beta = 0.32$, $p < 0.001$) and body surveillance ($\beta = 0.27$, $p < 0.001$); body surveillance is also positively associated with exercise for weight control ($\beta = 0.29$, $p < 0.001$). As the direct association between selfie-posting and exercise for weight control is observed to be significant, it holds that *body surveillance* partially mediates this association. The direct test on the mediating effect also indicates that the indirect effect is significant (mediating effect = 0.08, SE = 0.03, 95%CI [0.03, 0.14]; direct effect = 0.32, SE = 0.08, 95%CI [0.17, 0.47]). The results suggest that selfie-posting on SNSs is not only directly associated with weight control motivation for exercise among young women with a low BMI, but is also indirectly associated with weight control motivation for exercise through body surveillance in these same women.

4. Discussion

From the perspective of self-perception theory (Schienker et al., 1994) and objectification theory (Fredrickson & Roberts, 1997), this study examines the association between selfie-posting on SNSs and exercise for weight control among young Chinese women with a low BMI, as well as the mediating effect of body surveillance.

First, the findings show that selfie-posting on SNSs is positively associated with exercise for weight control among young women with a low BMI, suggesting that selfie-posting is a risk factor for unhealthy exercise motivations among such women. Although people usually edit and retouch selfies to make themselves look thinner and more attractive on SNSs (Cohen et al., 2018), this "photoshopping" may not only make women focus on their flaws or imperfections (Mills et al., 2018), but may also make them perceive actual-ideal discrepancies (Halpern, Katz, & Carril, 2017), leading to further body image dissatisfaction. Motivated by these factors, women often make efforts (e.g., exercise) to eliminate the perceived ideal gap (Furnham, Badmin, & Sneade, 2002). Thus, the process of selfie-editing may cause women to feel dissatisfied with their bodies which may, in turn, result in increased motivation to control their weight through exercise. In addition, the association between

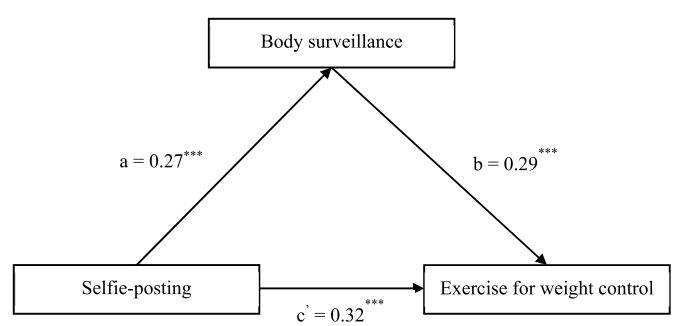

Fig. 1. Body surveillance as a mediator in the relationship between selfie-posting and exercise for weight control.
Note: Path values are path coefficients; ***$p < 0.001$.

Table 1
Descriptive statistics and correlations between variables.

Variables	$M \pm SD$	Correlations					
		1	2	3	4	5	6
1.Age	20.27 ± 1.31	1					
2.Body mass index (BMI)	19.66 ± 1.79	−0.05	1				
3.Number of friends on SNSs	201.46 ± 138.97	−0.08	−0.03	1			
4.Selfie-posting	2.90 ± 0.78	0.07	−0.04	0.16**	1		
5.Body surveillance	4.13 ± 0.81	−0.06	−0.01	0.09	0.26**	1	
6.Exercise for weight control	3.83 ± 1.32	−0.10*	0.31**	0.02	0.21**	0.22**	1

Note. $n = 467$, *$p < 0.05$, **$p < 0.01$.

selfie-posting and exercise for weight control is also consistent with the self-perception theory; that is, characteristics presented in selfies are able to influence individuals' self-perception and future behavior (motivation) (Niu et al., 2019 Schienker et al., 1994).

The results further uncovered one potential mechanism underlying this association by revealing the mediating role of body surveillance among young women with a low BMI. As objectification theory points out, the most common form of sexual objectification experience is the *evaluation of the body*, which is mainly externalized into objectification gaze and comments based on physical appearance (Fredrickson & Roberts, 1997). These sexual objectification experiences can make women internalize an observer's perspective, viewing their own body as evaluated objects, and frequently checking their physical appearance (Cohen et al., 2018). When taking selfies, individuals view what they look like in the pictures (Qiu et al., 2015). At the same time, such selfies can be viewed and commented on by other users, which may be also interpreted as affirmations of attractiveness (Butkowski et al., 2019). Thus, the selfie is considered as a new experience of sexual objectification, as enabled by the experience of cyberspace (Zheng et al., 2019), which can result in the development of body surveillance. In addition, underweight and normal weight young women usually have a tendency to misperceive themselves as being overweight (Fan, Jin, & Khubchandani, 2014; Kim & So, 2014 Lo et al., 2011), which may also contribute to greater self-scrutiny regarding their weight. The frequency of body checking magnifies the preoccupation with weight and shape (Shafran, Lee, Payne, & Fairburn, 2007), thereby further motivating women to take action (exercise) in order to control their weight. Thus, underweight and normal weight young women with high body surveillance tendencies tend to exercise more for weight control. In addition, by viewing such tendencies through the combination of self-perception theory and objectification theory, such tendencies suggest that self-presentation (selfie-posting) on SNSs would lead to corresponding behaviors (motivations, exercise for weight control) through one's cognition (body surveillance). That is, body surveillance mediates the association between selfie-posting and exercise for weight control among young women with a low BMI.

While these findings provide important information about the factors influencing weight control motivation for exercise among young women with a low BMI, there are several limitations. First, the cross-sectional design employed in this study cannot draw conclusions concerning causality. In particular, the causal relationship between selfie-posting and exercise for weight control remains unknown. Future studies can design experimental or longitudinal studies so as to further investigate the causality between these factors. Second, the types of selfies that women posted on SNSs may also influence their motivation to exercise; future research can, therefore, investigate the relationship between different types of selfies and exercise motivations in greater depth. Third, this study only focuses on women with BMI lower than 24 and their corresponding motivation to exercise for weight control. Future studies can include a greater number of participants with different BMI scores, focusing on exercise motivations other than appearance. Lastly, there may be other mechanisms (such as body dissatisfaction) exerting influence on the relationship between selfie-posting and exercise for weight control. Further research may explore such possible mechanisms so as to assist in the understanding of the relationship between selfie-posting and exercise motivations.

Despite these limitations, this study has several theoretical and practical implications. Theoretically, the findings shed light on factors underlying weight control motivation for engaging in exercise, and the impact of selfie behaviors among young women with a low BMI. This study's findings may also expand the self-perception theory into a new area by focusing on exercise, rather than eating. Practically, as one of the most common self-presentation behaviors on SNSs, selfie-posting found to be positively associated with maladaptive exercise motivations among young women with a low BMI; that is, exercising for appearance-related reasons (e.g., exercise for weight control). Thus, it is of great importance to guide young women to use SNSs reasonably and healthily. Young women should also be aware of the potentially negative influences of selfie-posting. For young women with a low BMI, in particular, it may be necessary to avoid engaging in selfie-behaviors on SNSs and to reduce the monitoring of one's own outward appearance, as well as engaging in exercise along reasons for health, and not appearance.

5. Conclusion

Selfie-posting is an important part of online self-presentation among young people. Unfortunately, this study finds that this popular phenomenon is positively associated with an unhealthy exercise motivation (exercise for weight control) among young women with a low BMI, and that body surveillance mediates this association. This indicates that young women with a low BMI, and who do not need to lose weight, should be aware of the influences of selfie-behavior and the accompanying body surveillance on their maladaptive exercise motivations.

CRediT authorship contribution statement

Liangshuang Yao: Conceptualization, Formal analysis, Writing - original draft. **Gengfeng Niu:** Writing - review & editing. **Xiaojun Sun:** Writing - review & editing, Supervision. **Changying Duan:** Data curation. **Yueli Zheng:** Writing - review & editing.

Declaration of competing interest

None.

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